(5) 1. a) Express the translation of $\mathbf{R}^3$ defined by $f(x,y,z) = (x+4, y-2, z-2)$ as a product of two reflections, and check your answer.

b) Express the rotation by $\frac{3\pi}{2}$ about $(1,-1)$ as a product of two reflections.

2 c) (i) Show that if $P = (0,0)$, $Q = (1,1)$, $R = (1,-1)$ and $S = (1,-3)$, then there is an orientation preserving similarity $f : \mathbf{R}^2 \to \mathbf{R}^2$ such that $f(\overline{PQ}) = \overline{RS}$.

1 (ii) Find an orientation *reversing* similarity g, such that $g(\overline{PQ}) = \overline{RS}$.

a) $f(v) = v + (4,-2,-2)$. Let $H_0 = \{v \in \mathbb{R}^3 / v \cdot (2,-1,-1) = 0\}$,

and $H_1 = \{v \in \mathbb{R}^3 / v \cdot (2,-1,-1) = \frac{(4,-2,-2)}{2} \cdot (2,-1,-1) = 6\}$. Then,

$f = R_{H_1} R_{H_0}$, $\quad R_{H_0}(v) = v - \frac{2((2,-1,-1)\cdot v)}{6}(2,-1,-1)$

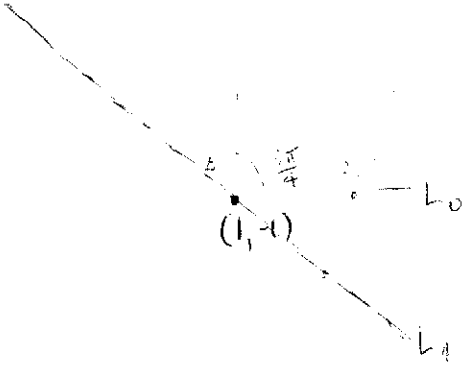
and $R_{H_1}(w) = w + \frac{2(6 - (2,-1,-1)\cdot w)}{6}(2,-1,-1)$ ∴ if we let $n = (2,-1,-1)$,

$R_{H_1} R_{H_0}(v) = R_{H_1}\left(v - \frac{(n\cdot v)n}{3}\right) = v - \frac{1}{3}(n\cdot v)n + \frac{1}{3}\left(6 - n\cdot\left(v - \frac{1}{3}(n\cdot v)n\right)\right)n$

$= v - \frac{1}{3}(n\cdot v)n + \frac{1}{3}\left(6 - n\cdot v + \frac{1}{3}(n\cdot v)\frac{\|n\|^2}{6}\right)n = v - \frac{1}{3}(n\cdot v)n + \frac{1}{3}(6 + (n\cdot v))n$

$= v + 2n = v + (4,-2,-2)$, as required.

b)



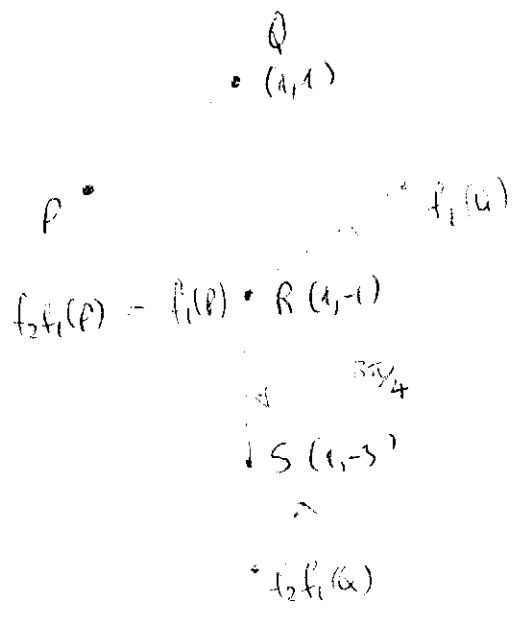
Let $L_0 = \{(x,-1) \mid x \in \mathbb{R}\}$

and $L_1 = \{(x,-x) \mid x \in \mathbb{R}\}$.

Then $R_{L_1} R_{L_0}$ is rotation about

$L_0 \cap L_1 = \{(1,-1)\}$ by $2 \cdot \frac{3\pi}{4} = \frac{3\pi}{2}$.

c) (i)



Let $f_1(v) = v + (1,-1)$.

$f_2(v) = R_{(1,-1)(-\frac{3\pi}{4})}$.

Then, $f_2f_1(Q) = f_2(2,0)$

$= (1,-1-\sqrt{2})$

Now let $f_3(v) = \sqrt{2}(v-R) + R$

Note that $f_3(R) = R$.

Then,

$f_3(1,-1-\sqrt{2}) = \sqrt{2}(0,-\sqrt{2}) + (1,-1) = (0,-2) + (1,-1) = (1,-3)$

$= S$

Hence, if $g = f_3f_2f_1$, then $g(P) = R$ and

$g(Q) = f_3(1,-1-\sqrt{2}) = S$.

Since g is a composition of (orientation preserving) similarities, g is an orientation preserving similarity, and $g(\overline{PQ}) = \overline{RS}$.

(ii) If we reflect by a reflection in the line L through R and S, then $g' = R_L g$ satisfies

$g'(\overline{PQ}) = \overline{RS}$ (since R_L fixes $\overline{RS}$)

and g' is orientation reversing, because R_L, and g preserves orientation

2. Let $f : \mathbf{R}^2 \to \mathbf{R}^2$ be defined by

$$f(x, y) = \frac{1}{5}(10 + 4x - 3y, -3x - 4y).$$

a) Show that f is an orientation reversing isometry of $\mathbf{R}^2$.

b) Find a line $L \subset \mathbf{R}^2$, and a vector w which is parallel to L, so that f is reflection in L followed by translation by w.

c) Check that f is the same as the isometry which is translation by w followed by reflection in L.

a) $f(v) = \underbrace{\frac{1}{5}\begin{bmatrix} 4 & -3 \\ -3 & -4 \end{bmatrix}}_{A} v + \begin{bmatrix} 2 \\ 0 \end{bmatrix}$; Now, $AA^t = \frac{1}{25}\begin{bmatrix} 25 & 0 \\ 0 & 25 \end{bmatrix} = I_2$, so $A \in O(2)$.

Hence, f is an isometry. (2)

Since $\det A = \frac{1}{25} \cdot (-25) = -1$, f reverses orientation. (1)

b) If $f(v) = w + R_L(v)$, then $f(f(0)) = 2w$. But $f(f(0))$

$$= \frac{1}{5}\begin{bmatrix} +4 & -3 \\ -3 & -4 \end{bmatrix}\begin{pmatrix} 2 \\ 0 \end{pmatrix} + \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \frac{1}{5}\begin{bmatrix} 8 \\ -6 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{pmatrix} 18/5 \\ -6/5 \end{pmatrix} \quad w = \begin{bmatrix} 9/5 \\ -3/5 \end{bmatrix}$$

We can find L now by solving $f(v) = v + w$, so this is

$$\frac{1}{5}\begin{bmatrix} 4 & -3 \\ -3 & -4 \end{bmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 9/5 \\ -3/5 \end{bmatrix} \quad \text{i.e.}$$

$$\frac{1}{5}\left[\begin{array}{cc|c} -1 & -3 & -1 \\ -3 & -9 & 3 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 3 & 1 \\ 0 & 0 & 0 \end{array}\right] \qquad \begin{aligned} x &= 1 - 3\lambda \\ y &= \lambda \end{aligned}$$

Hence $L = \{(x, y) \mid x + 3y = 1\}$. (1)

c) Let $g(v) = R_L(v+w)$; $R_L(u) = u + \frac{2(1-(1,3)\cdot u)}{10}(1,3)$

$= v + w + \frac{2(1-(v+w)\cdot(1,3))}{10}(1,3)$; (but $w\cdot(1,3) = 0$)

$= v + \frac{2(1-v\cdot(1,3))}{10}(1,3) + w$

$= R_L(v) + w = f(v)$, as required.

n_1 n_2

3. Let $C_0 = \{v \in \mathbf{S}^2 \mid (1,-1,0)\cdot v = 0\}$ and $C_1 = \{v \in \mathbf{S}^2 \mid (-1,0,1)\cdot v = 0\}$ be great circles on $\mathbf{S}^2$.

3 a) Find $C_0 \cap C_1$.

3 b) Find formulae for the reflections R_{C_0} and R_{C_1} in C_0 and C_1 respectively.

3 c) Use your answers from (b) to find $A \in O(3)$ so that $R_{C_1} \circ R_{C_0} = Av$.

1 d) Use the formula $R_{a,\theta}(v) = (v\cdot a)(1-\cos\theta)a + (\cos\theta)v + (\sin\theta)(a \times v)$ to verify that the $R_{C_1} \circ R_{C_0}$ is the rotation by $\frac{2\pi}{3}$ about the point $\frac{\sqrt{3}}{3}(1,1,1) \in \mathbf{S}^2$. (Don't forget that the formula is valid only if $||a|| = 1$.)

a) $C_0 \cap C_1 = \{v \in S^2 / (1,-1,0)\cdot v = (-1,0,1)\cdot v = 0\} = \{v \in S^2 / v = \lambda(1,-1,0)\times(-1,0,1), \lambda \in \mathbb{R}\}$

$\begin{vmatrix} i & j & k \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = (-1,-1,-1)$. Hence $C_0 \cap C_1 = \left\{\pm \frac{\sqrt{3}}{3}(1,1,1)\right\}$.

b) $R_{C_0}(v) = v - \frac{2(n_1\cdot v)}{2} n_1 = v - (n_1\cdot v)n_1$; $R_{C_1}(v) = v - (n_2\cdot v)n_2 = (3,y,x)$
$= (y,x,3)$

c) Then $R_{C_1}R_{C_0}(v) = R_{C_1}(v - (n_1\cdot v)n_1) = v - (n_1\cdot v)n_1 - (n_2\cdot(v-(n_1\cdot v)n_1)n_2$

$= v - (n_1\cdot v)n_1 - (n_2\cdot v)n_2 + (n_1\cdot v)(n_1\cdot n_2)n_2 = v - (n_1\cdot v)n_1 - (n_2\cdot v)n_2 - (n_1\cdot v)n_2$

$\therefore R_{C_1C_0}(x,y,3) = (x,y,3) - (x-y)(1,-1,0) - (3-x)(-1,0,1) - (x-y)(-1,0,1)$

$= (x-x+y+3-x+x-y,\ y+x-y,\ 3-3+x-x+y)$

$= (3,x,y)$

$= \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}\begin{pmatrix} x \\ y \\ 3 \end{pmatrix}$. So $A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$.

d) If $a = \frac{\sqrt{3}}{3}(1,1,1)$ and $\theta = 2\pi/3$, then $\cos\theta = -\frac{1}{2}$, $\sin\theta = \frac{\sqrt{3}}{2}$.

Hence, $R_{a,\theta}(v) = (v \cdot a)\left(\frac{3}{2}\right)a - \frac{1}{2}v + \frac{\sqrt{3}}{2}\, a \times v.$

$$= \frac{1}{2}(v \cdot (1,1,1))(1,1,1) - \frac{1}{2}v + \frac{1}{2}\,(1,1,1) \times v$$

$$\begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ x & y & 3 \end{vmatrix} = (3-y, -(3-x), y-x)$$

$$\therefore R_{a,\theta}(v) = \frac{1}{2}(x+y+3)(1,1,1) - \frac{1}{2}(x,y,3) + \frac{1}{2}(3-y, x-3, y-x)$$

$$= \frac{1}{2}(x+y+3-x+3-y,\ x+y+3-y+x-3,\ x+y+3-3+y-x)$$

$$= \frac{1}{2}(2z, 2x, 2y) = (z, x, y) = R_{C_1} \circ R_{C_0}(v).$$

4. Let $g : \mathbf{R}^3 \to \mathbf{R}^3$ be defined by $g(v) = Av$ where $A = \frac{1}{3}\begin{bmatrix} -2 & -2 & 1 \\ 1 & -2 & -2 \\ -2 & 1 & -2 \end{bmatrix}$.

a) Show that g is an orientation reversing isometry of $\mathbf{R}^3$.

b) Find a vector $a \in \mathbf{S}^2$, and $\theta \in [0, 2\pi)$ such that g is rotation by θ about a, followed by reflection in the plane through the origin perpendicular to a.

c) Check your answer in (b) by an explicit computation.

a) Since $AA^t = \frac{1}{9}\begin{bmatrix} -2 & -2 & 1 \\ 1 & -2 & -2 \\ -2 & 1 & -2 \end{bmatrix}\begin{bmatrix} -2 & 1 & -2 \\ -2 & -2 & 1 \\ 1 & -2 & -2 \end{bmatrix} = \frac{1}{9}\begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = I_3$, $A \in O(3)$.

Moreover, $\det A = \frac{1}{27}\begin{vmatrix} -2 & -2 & 1 \\ 1 & -2 & -2 \\ -2 & 1 & -2 \end{vmatrix} = \frac{1}{27}\begin{vmatrix} 0 & -6 & -3 \\ 1 & -2 & -2 \\ 0 & -3 & -6 \end{vmatrix} = -\frac{1}{27}\begin{vmatrix} -6 & -3 \\ -3 & -6 \end{vmatrix} = -1$, so

g reverses orientation.

b) We solve $Aa = -a$: $[A + I_3 \mid 0] = \frac{1}{3}\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 1 & 1 & -2 & 0 \\ -2 & 1 & 1 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 3 & -3 & 0 \\ 0 & -3 & 3 & 0 \end{array}\right]$

$\sim \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$ $\therefore \begin{matrix} x = s \\ y = s \\ z = s \end{matrix}$ $s \in \mathbb{R}$. So we can choose $a = \frac{\sqrt{3}}{3}(1,1,1)$.

To find θ, choose $w \in H = \{v \in \mathbb{R}^3 / v \cdot a = 0\}$; we take $w = (1,-1,0)$

Then, $g(w) = \frac{1}{3}\begin{bmatrix} -2 & -2 & 1 \\ 1 & -2 & -2 \\ -2 & 1 & -2 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 0 \\ 3 \\ -3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$. So, $\frac{w \cdot g(w)}{\|w\| \cdot \|g(w)\|} = \frac{-1}{2}$,

so $\theta = 2\pi/3$ or $4\pi/3$. To determine which, consider $(w \times g(w)) = \begin{vmatrix} i & j & k \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{vmatrix}$

$= (1,-(-1),1)$, which is a positive multiple of a. Hence $\sin\theta > 0$ and so

$\theta = 2\pi/3$

c). From Q3, we already know that

$$R_{a,0}(v) = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} v. \quad \text{ie.}$$

The reflection in H is $R_H(v) = v - \frac{2}{3}((1,1,1)\cdot v)(1,1,1)$

$$\therefore R_H(x,y,z) = (x,y,z) - \tfrac{2}{3}(x+y+z,\ x+y+z,\ x+y+z)$$

$$= \tfrac{1}{3}(x-2y-2z, -2x+y-2z, -2x-2y+z).$$

Therefore, $R_H R_{a,0}(v) = R_H(z,x,y)$

$$= \tfrac{1}{3}\begin{bmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ -2 & -2 & 1 \end{bmatrix}\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= \tfrac{1}{3}\begin{bmatrix} -2 & -2 & 1 \\ 1 & -2 & -2 \\ -2 & 1 & -2 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = q(v), \text{ as required.}$$

5. The latitude and longitude of Ottawa and Madrid are given below:

	Latitude	Longitude
Madrid	$40°26' N$	$3°42' W$
Ottawa	$45°24' N$	$75°43' W$

(Round all answers to 3 decimal places.)

(2) a) Find the (ϕ, θ) coordinates of Ottawa and Madrid from the given data. Express your answer in radians.

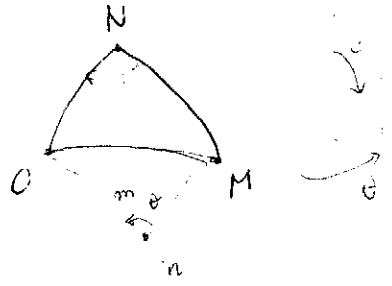
(4) b) Find the shortest distance from Ottawa to Madrid. Assume that the earth is a sphere of radius 6380 km.

(2) c) Find the bearing of Madrid from Ottawa. Express your answer in radians.

(1) d) If you were to start at Ottawa and maintain the bearing found in (c), where would you end up?

a) $40°26' \approx 40.433° \approx .706$ radians
$3°42' \approx 3.7° \approx .065$ radians

$45°24' = 45.4° \approx .792$ radians
$75°43' \approx 75.717° \approx 1.322$ radians



$\therefore \phi(\text{Madrid}) \approx \pi/2 - .706 \approx .865 (\approx 49°34')$ ½
$\theta(\text{Madrid}) \approx 2\pi - .065 \approx 6.219 (\approx 356°18')$ ½
½ $\phi(\text{Ottawa}) \approx \pi/2 - .792 \approx .778 (\approx 44°36')$ ½
$\theta(\text{Ottawa}) \approx 2\pi - 1.322 \approx 4.962 (\approx 284°17')$ ½

b) By the cosine rule, $\cos n = \cos\theta\cos m + \sin\theta \sin m \cos \angle N$ (1)

But (1) $\angle N = \theta(\text{Madrid}) - \theta(\text{Ottawa}) \approx 1.257$, while $\theta = \phi(M)$, $m = \phi(\text{Ottawa})$ ½

$\therefore \cos n \approx .4618 + .1650 \approx .6268 \therefore n \approx .8934$ rad $(\approx 51°11')$ ½ the distance from Ottawa to Madrid along a great circle arc is

$\approx 6380 \times .8934 \approx 5700$ km. (1)

½ [illegible]

c) To find $\angle O$, we again use the cosine rule:

(1/2) $\cos\theta = \cos m \cos n + \sin m \sin n \cos\angle O$

$\therefore \cos\angle O = \dfrac{\cos\theta - \cos m \cos n}{\sin m \sin n} \approx \dfrac{.6486 - .4463}{.5471} \approx .3697$

$\therefore \angle O \approx 1.192$ radian (1/2) $(< \pi/2)$
$(\approx 68^\circ 18')$
∴ bearing of Madrid from Ottawa is

(1/2) (1/2) $N\ 1.192\ E$ $(\approx N\ 68^\circ 18' E)$
(constant) (rad.)
(One initially heads a little north, but Madrid is South of Ottawa!)

d) If we keep bearing north, we will eventually (1) end up at the north pole.

6. Let $N = (0, 0, 1)$ and f and g denote the stereogrpahic and the central projections

$$f : \mathbf{S}^2 - \{(0, 0, 1)\} \to H_0 = \{(x, y, 0) \mid x, y \in \mathbf{R}^2\}$$

and

$$g : \mathbf{S}^2_- = \{(x, y, z) \in \mathbf{S}^2 \mid z < 0\} \to H_{-1} = \{(x, y, -1) \mid x, y \in \mathbf{R}^2\}$$

and let

$$C = \{(x, y, z) \in \mathbf{S}^2 \mid x + y = 0 \text{ and } z < 0\}$$

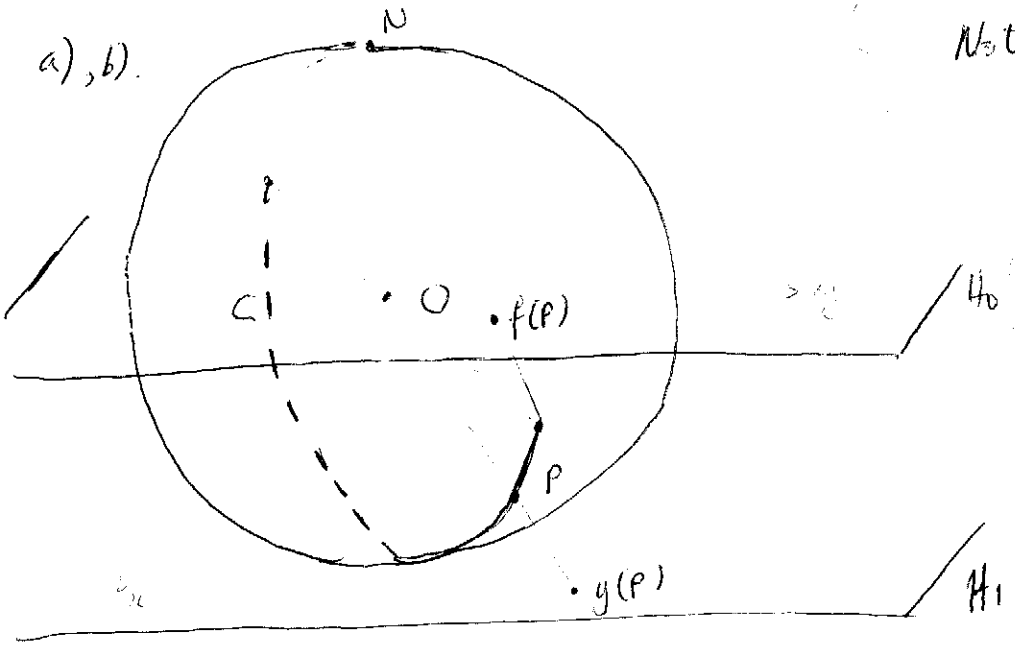
be part of a great circle on $\mathbf{S}^2$.

a) Show that $f(C)$ is a line in H_0 and that $g(C)$ is a line in H_{-1}.

b) Show that $f(C)$ and $g(C)$ are parallel.

c) Will $f(C')$ be a line in H_0 for all great circles C' in $\mathbf{S}^2$?

d) Explain why there can be no isometry from any region of $\mathbf{S}^2$ (with non-zero area) into the plane $\mathbf{R}^2$.



a), b).

Note that $C \subset H_2 = \{(x, y, 3) \mid x + y = 0\}$, which is a plane in $\mathbb{R}^3$ passing through the origin O and N.

Thus, if $P \in C$, then $P \in H_2$, so $f(P) \in H_0 \cap \overline{NP} \subset H_0 \cap H_2$, by def^n

since N and $P \in H_2$.

Moreover, $g(P) \in H_1 \cap \overrightarrow{OP} \subset H_1 \cap H_2$, since $O, P \in H_2$.

Thus $f(C) \subset H_0 \cap H_2$, which is a line in H_0, and $g(C) \subset H_1 \cap H_2$, which is a line in H_1. But $H_0 \parallel H_1$, so $H_0 \cap H_2 \parallel H_1 \cap H_2$, i.e.

$f(C)$ and $g(C)$ lie in parallel lines

c) No. Let $C' = H_0 \cap S^2$. Then for all $P \in C'$, $f(P) = P$, so that $f(C') = C'$, which does not lie in a line in H_0!

d). Any isometry of a region of non-zero area in S^2 to $\mathbb{R}^2$ will in particular be an isometry of a spherical triangle (of non-zero area) in S^2 to some Euclidean triangle in $\mathbb{R}^2$. Isometry preserves angles, but the angle sum of the spherical triangle exceeds π, while that of the Euclidean triangle is exactly π. Therfore, no such isometry exists.

7. Suppose that $f : \mathbf{R}^2 \to \mathbf{R}^2$ is defined by $f(v) = Ev + a$, where E is an invertible 2×2 matrix and $a \in \mathbf{R}^2$. (We do *not* assume that E is orthogonal.)

a) Show that f preserves geodesics (i.e. takes lines to lines).

b) Suppose that f is conformal, i.e. f preserves angles. Show that f is a similarity.

c) Suppose that f is conformal, and that f also preserves area. Show that f is an isometry.

a) Let $L = \{ v_0 + tv_1 \mid t \in \mathbb{R} \}$ for some $v_0 \in \mathbb{R}^2$, $v_1 \neq 0$, $v_1 \in \mathbb{R}^2$. Then,

$f(L) = \{ E(v_0 + tv_1) \mid t \in \mathbb{R} \} = \{ (Ev_0 + a) + t(Ev_1) \mid t \in \mathbb{R} \}$. Let

$v_1' = Ev_1 (\neq 0$ because E is invertible$)$, $v_0' = Ev_0 + a$. Then

$f(L) = \{ v_0' + tv_1' \mid t \in \mathbb{R} \}$ is a line in $\mathbb{R}^2$.

b) If $P, Q, R \in \mathbb{R}^2$, then $\dfrac{(f(P) - f(Q)) \cdot (f(R) - f(Q))}{\|f(P) - f(Q)\| \, \|f(R) - f(Q)\|} = \dfrac{(P - Q) \cdot R - Q}{\|P - Q\| \, \|R - Q\|}$

for all P, Q, R. Hence, $\dfrac{Ev \cdot Ew}{\|Ev\| \, \|Ew\|} = \dfrac{v \cdot w}{\|v\| \cdot \|w\|}$ $(*)$ for all $v, w \in \mathbb{R}^2$.

Let $e_1' = Ee_1$, $e_2' = Ee_2$. Then $e_1' \cdot e_2' = 0$; and

$\|e_1'\| = k_1$ and $\|e_2'\| = k_2$ for some $k_1, k_2 > 0$. We show

$k_1 = k_2$ as follows: Let $v = e_1$, $w = e_1 + e_2$. Then $(*)$ implies

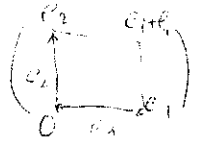
$$\frac{e_1' \cdot (e_1' + e_2')}{k_1 \sqrt{k_1^2 + k_2^2}} = \frac{e_1 \cdot (e_1 + e_2)}{1 \cdot \sqrt{2}} \quad \text{i.e.} \quad k_1^2 = \frac{k_1 \sqrt{k_1^2 + k_2^2}}{\sqrt{2}}$$

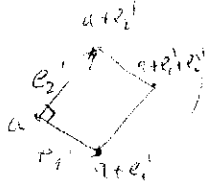
i.e. $2k_1^4 = k_1^2(k_1^2 + k_2^2)$ $\therefore k_1^2 k_2^2 = k_1^4$ so $k_1^2 = k_2^2$, so $k_1 = k_2$.

Hence $EE^t = k_1^2 \cdot I_2$, so $E = k_1 A$, where $A \in O(2)$. Hence,

f is a similarity.

(c) by (b), we know $E = \xi A$ where $A \in O(2)$.

If f preserves area, then $1 = \text{area}$ 

$= \text{area}$  $= \|e_1'\| \|e_2'\| = \xi_1^2$.

Hence $\xi_1 = \pm 1$. Thus, $E = A$ or $-A$, so E is orthogonal.

Thus, f is an isometry. ∎